



Computer Networks & Operation Systems

Department of Computer Education and Instructional
Technology

Grup 5
NetQues 

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GitHub Repo Link: <https://github.com/NurullahDAGHAN/OS-Project-xMeine.git>

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Representation: <https://drive.google.com/file/d/1Wcd1ZCC2oQMuLVv0GfkoUi3pO9Kb5mWq/view?usp=drivesdk>

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Network Cable Demo: Educational Computer Networks Game

1. Project Overview and Vision

NetQues is a next-generation EdTech (Educational Technology) project designed to teach the fundamentals of computer networks and operating systems to beginner students through a gamified mobile simulation. Instead of presenting network topics as abstract definitions, it aims for long-lasting learning by turning familiar daily situations—such as connecting a computer to a modem or choosing an IP address—into interactive tasks. The core vision is to build a sequence of short levels that teach computer networks step-by-step through actions, visual cause-and-effect, and immediate feedback.

2. Visual Identity, User Interface, and Interaction Design (UI/UX)

The project is specifically designed to reduce students' cognitive load and increase focus.

- **Dynamic Screen and Orientation Adaptation:** The application is built as a portrait-first, hyper-casual mobile experience. However, utilizing the flexible structure of Flutter and the Flame engine, it dynamically adapts to device screen rotations (landscape/portrait) and various screen resolutions. The isometric 3D game scene consistently remains centered.
- **Minimalist and Accessible Design:** The user-friendly interface, featuring pastel green and cream tones, provides a clean login screen and instant language selection (Turkish/English), ensuring a frictionless user experience at global standards.
- **Isometric Drag-and-Drop Mechanics:** Interactions are not performed via static buttons, but by dragging physical objects in the game area (e.g., an Ethernet cable or an IP card) and dropping them onto the target (modem/router/computer).

3. Pedagogical Approach and Gamification

The learning process is built upon concrete examples, instant feedback, and the educational principle of "action before explanation."

- **Progressive Disclosure (Step-by-Step Unlocking):** The levels are interconnected. A new level is not unlocked until the student successfully completes the previous one, ensuring that the knowledge is fully absorbed before moving forward.
- **Advanced Profile and Motivation System:** A Daily Login Streak system is integrated to help students build a consistent learning habit. The in-app "Profile" screen tracks completed levels, earned achievement badges, and a 7-day activity calendar to keep student motivation consistently high.

4. Playable Levels and Learning Outcomes

The application consists of a progressive sequence of 7 levels. Each level gamifies a specific and sequential networking concept:

- **Level 1 (Physical Connectivity - Ethernet):** The student drags the Ethernet cable plug to the correct modem port. If dropped in the wrong place, the cable returns to a safe state with a short hint; if connected correctly, a physical connection is established.
 - *Learning Outcome:* The student understands that internet access begins with a physical connection between the computer and the modem/router.
- **Level 2 (Logical Addressing - IP Address):** The modem is represented as being on the 192.168.1.x network. The student must choose the correct IP address (192.168.1.24) from multiple options and assign it to the computer.
 - *Learning Outcome:* The student learns that devices on the same local network must use compatible IP addresses to communicate.
- **Level 3 (External Routing - Default Gateway):** To communicate outside the local network, the student must identify and drag the correct gateway device card (192.168.1.1 (Modem)) to the computer, ignoring distractors like the DNS or the PC itself.
 - *Learning Outcome:* The student learns that a "Default Gateway" acts as the designated exit point for traffic leaving the local network to reach the broader internet.
- **Level 4 (Name Resolution - DNS Server):** The computer attempts to reach a website (e.g., *example.com*). The student must select the "DNS Server" card to translate the domain name into an IP address.
 - *Learning Outcome:* The student grasps the fundamental role of a DNS server as the "phonebook" of the internet.
- **Level 5 (Network Boundaries - Subnet Mask):** The game prompts the user to define the network boundaries. The student must select the correct subnet mask (255.255.255.0) for a standard local network.
 - *Learning Outcome:* The student understands that a subnet mask defines the boundaries of a local network, separating the network portion of an IP address from the host portion.
- **Level 6 (Automatic Configuration - DHCP Service):** As the network grows, manual IP configuration becomes difficult. The student must identify and drop the "DHCP Server" card onto the system to automate the IP leasing process.
 - *Learning Outcome:* The student learns that the DHCP service automates the assignment of IP addresses, subnet masks, and default gateways to devices on a network.
- **Level 7 (Network Security - Firewall):** The student configures basic security rules by selecting the correct traffic protocol to permit safe web browsing while blocking unwanted connections.
 - *Learning Outcome:* The student gains a basic understanding of how firewalls filter incoming and outgoing network traffic based on predetermined security rules.

5. Technical Architecture: Flutter and Flame Integration

The project is built with a modular approach that separates the data, game, and user interface (UI) layers.

- **Flutter:** Used for the mobile application shell, UI panels, heads-up displays (HUD), dialogs, profile pages, and cross-platform builds.
- **Flame Game Engine:** Written in Dart, this engine is responsible for the game loop, isometric scene rendering, and managing the physical drag interactions.
- State management is provided by *ValueNotifier*, which acts as a lightweight UI and game state manager for the prototype.

6. Offline-First Database Architecture: SQLite

A highly robust technical aspect of the project is its **SQLite local database architecture**, which allows the application to run smoothly and effectively without requiring an internet connection (offline-first). This ensures accessibility in schools with limited internet infrastructure.

- **levels & level_order:** These tables store level definitions, level order, titles, and educational messages.
- **level_objects & Task Tables (cable_goals, ip_selection_goals):** These store the types of objects in the game scene, task types, and target positions.
- **player_progress:** This table is the core of the application's gamification engine. It stores the unlock status, completion state, attempt counts, and hint usage locally and securely on the device. This structure completely isolates the database layer from the Flame game scene. The game scene receives pre-loaded level data from a *Repository* layer instead of querying SQLite directly, ensuring high performance.

Screenshots





